MIT | **Academy of Engineering**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Subject (Code):
Essentials of Data Science
(2304102L)

EDS CodeTantra Programs

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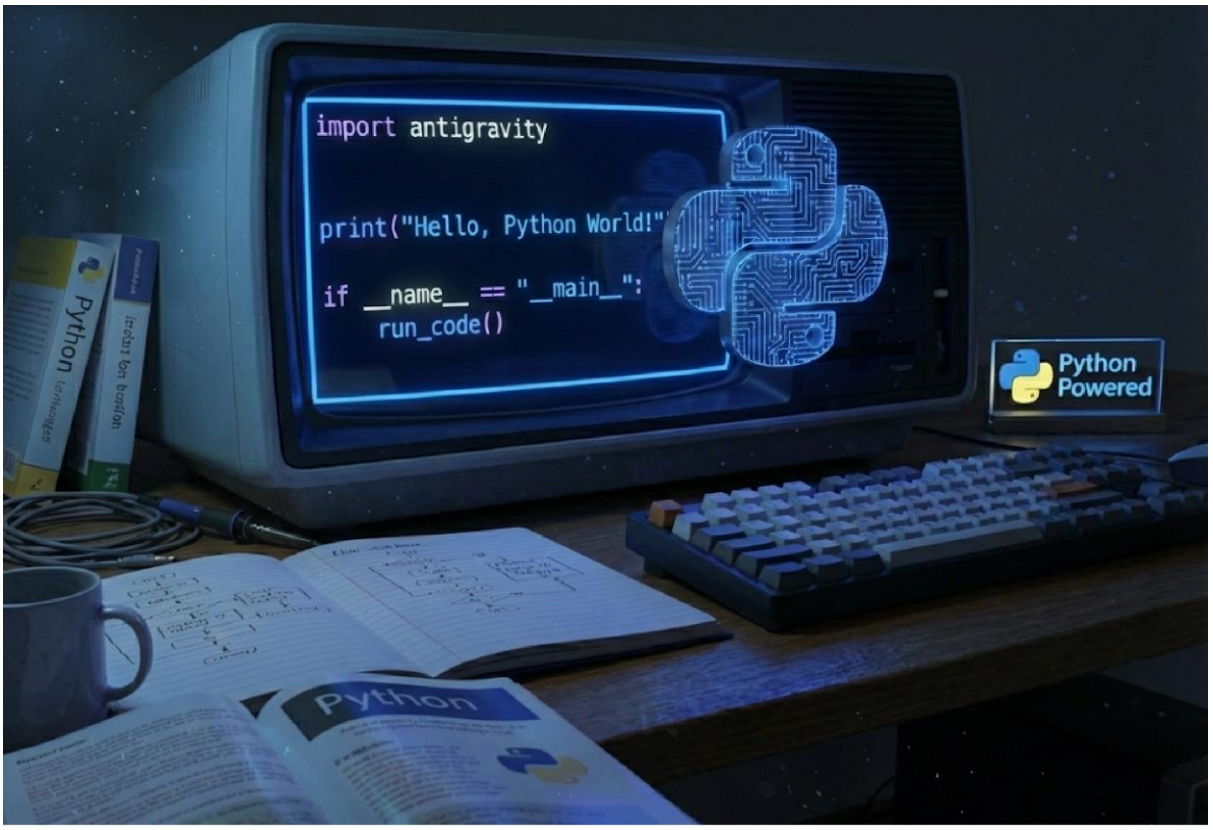


Practical 03

3.1 Practice Lab Assignments

3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, *r* and *c*, representing the number of rows and the number of columns.
- The next *r* lines contain *c* space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

```
import numpy as np

rows, cols= map(int,input().split())

elements = []

for _ in range(rows):
```

```
        elements.extend(map(int,input().split()))



array=np.array(elements).reshape(rows,cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)
```

Average time
0.012 s
12.42 ms

Maximum time
0.023 s
23.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 23 ms DEBUG

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]

Terminal Test cases

Test case 2 4 ms DEBUG

Expected output	Actual output
0 0	0 0
[]	[]
2	2
(0, 0)	(0, 0)
0	0

3.2 Lab Assignments

3.2.1. NumPy: Matrix Operations

The given code takes two *3×3* matrices, *matrix_a*, and *matrix_b*, as input from the user and converts them into NumPy arrays.

Task: You are required to compute and display the results of the following matrix operations:

1. **Addition** (*matrix_a* + *matrix_b*)
2. **Subtraction** (*matrix_a* - *matrix_b*)
3. **Element-wise Multiplication** (*matrix_a* * *matrix_b*)
4. **Matrix Multiplication** (*matrix_a* • *matrix_b*)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for *matrix_a*, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for *matrix_b*, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np

# Input matrices

print("Enter Matrix A:")

matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")

matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition

addition_result=matrix_a+matrix_b

print("Addition (A + B):")

print(addition_result)

# Subtraction

substraction_result=matrix_a-matrix_b

print("Subtraction (A - B):")

print(substraction_result)

# Multiplication (element-wise)

elementwise_multiplication_result=matrix_a*matrix_b

print("Element-wise Multiplication (A * B):")

print(elementwise_multiplication_result)

# Matrix multiplication (dot product)

matrix_multiplication_result=np.dot(matrix_a, matrix_b)

print("A dot B:")

print(matrix_multiplication_result)

# Transpose

transpose_a=matrix_a.T

print("Transpose of A:")

print(transpose_a)
```

Average time	Maximum time
0.054 s	0.064 s
54.00 ms	64.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 64 ms DEBUG

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B):	Addition (A + B):
[[2 3 4]	[[2 3 4]
[5 6 7]	[5 6 7]
[8 9 10]]	[8 9 10]]
Subtraction (A - B):	Subtraction (A - B):
[[0 1 2]	[[0 1 2]
[3 4 5]	[3 4 5]
[6 7 8]]	[6 7 8]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]]	[7 8 9]]
A dot B:	A dot B:
[[6 6 6]	[[6 6 6]
[15 15 15]	[15 15 15]
[24 24 24]]	[24 24 24]]
Transpose of A:	Transpose of A:
[[1 4 7]	[[1 4 7]
[2 5 8]	[2 5 8]
[3 6 9]]	[3 6 9]]

3.2.2. NumPy Horizontal and Vertical Stacking of Arrays

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

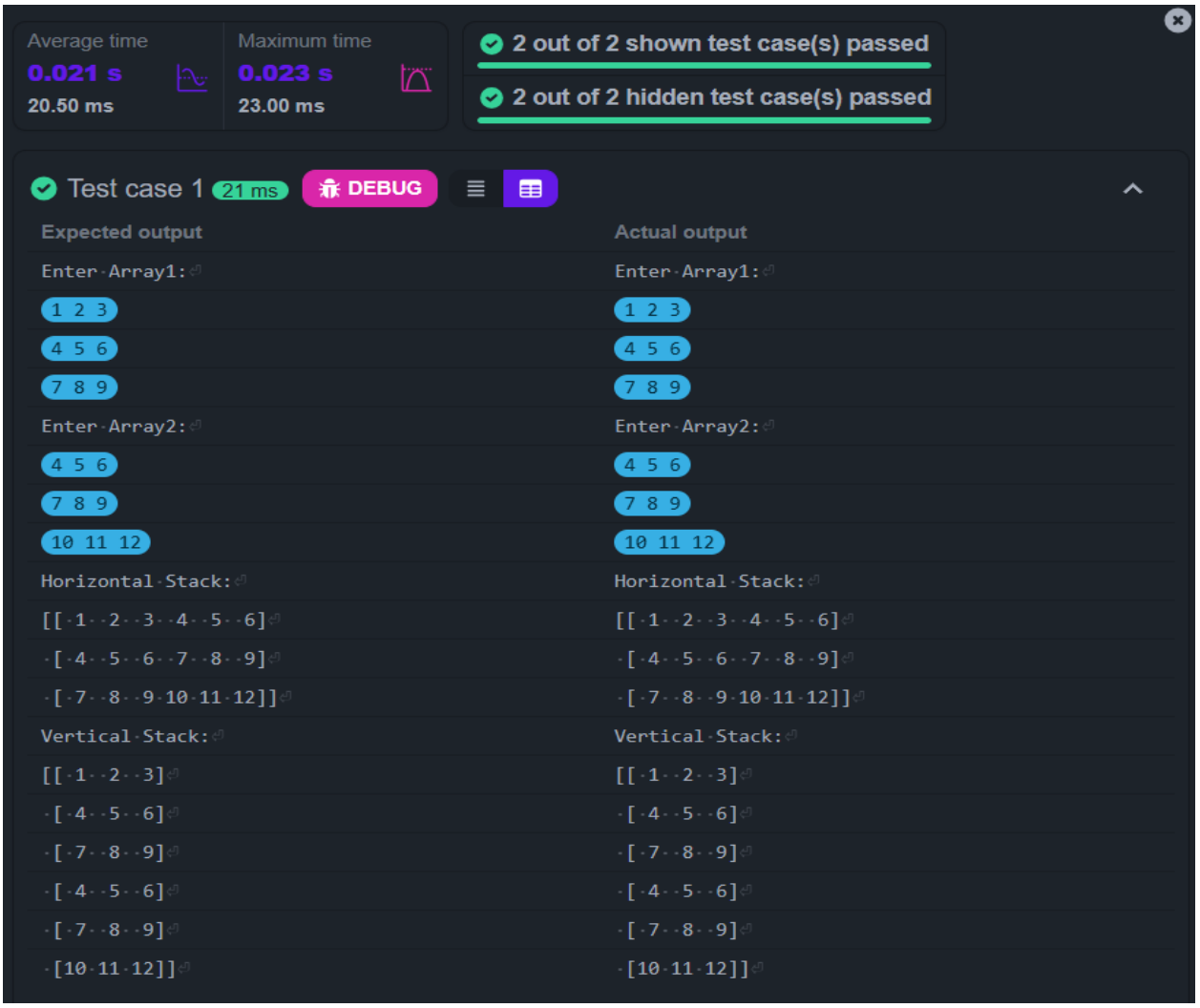
- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

# Input matrices

print("Enter Array1:")

arr1 = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Array2:")

arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)

a=np.hstack((arr1,arr2))

print("Horizontal Stack:")

print(a)

# Perform vertical stacking (vstack)

b=np.vstack((arr1,arr2))

print("Vertical Stack:")

print(b)
```



3.2.3. NumPy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.

- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

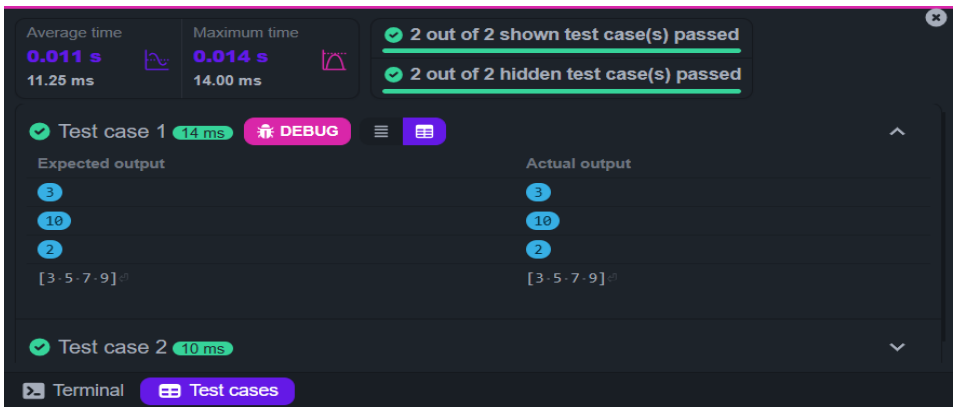
Input Format:

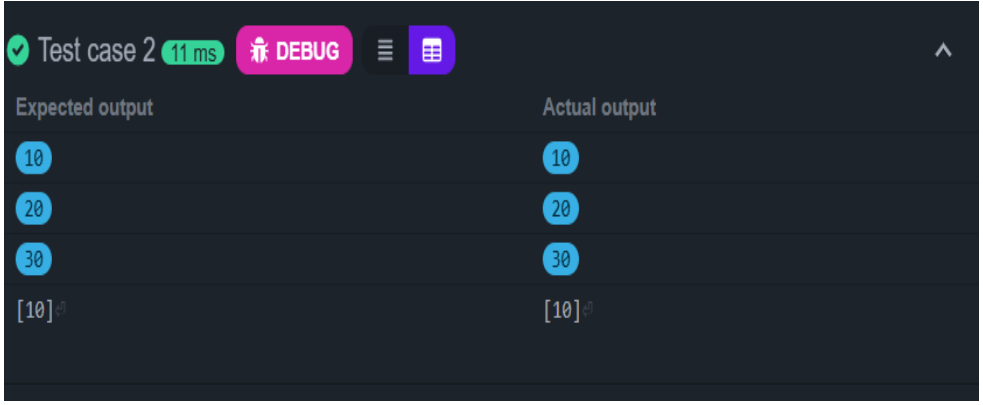
- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np

start = int(input())
stop = int(input())
step = int(input())

arr = np.arange(start,stop,step)

print(arr)
```





3.2.4. NumPy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. **Arithmetic Operations**:
   - Compute the element-wise sum, difference, and product of the two arrays.

2. **Statistical Operations**:
   - Calculate the mean, median, and standard deviation of array A.

3. **Bitwise Operations**:
   - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np

def array_operations(A, B):

    A=np.array(A)

    B=np.array(B)



    sum_result = A+B

    diff_result = A-B

    prod_result = A*B



    mean_A = np.mean(A)

    median_A = np.median(A)

    std_dev_A = np.std(A)



    and_result = A & B

    or_result = A | B

    xor_result = A ^ B
```

```
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))

    print("Element-wise Difference:", ' '.join(map(str, diff_result)))

    print("Element-wise Product:", ' '.join(map(str, prod_result)))



    print(f"Mean of A: {mean_A}")

    print(f"Median of A: {median_A}")

    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))

    print("Bitwise OR:", ' '.join(map(str, or_result)))

    print("Bitwise XOR:", ' '.join(map(str, xor_result)))



A = list(map(int, input().split()))  # Elements of array A

B = list(map(int, input().split()))  # Elements of array B

array_operations(A, B)
```

3.2.5. NumPy: Copying and Viewing Array

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the **original_array** and assigning it to **view_array**.
- Creating a copy of the **original_array** and assigning it to **copy_array**.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

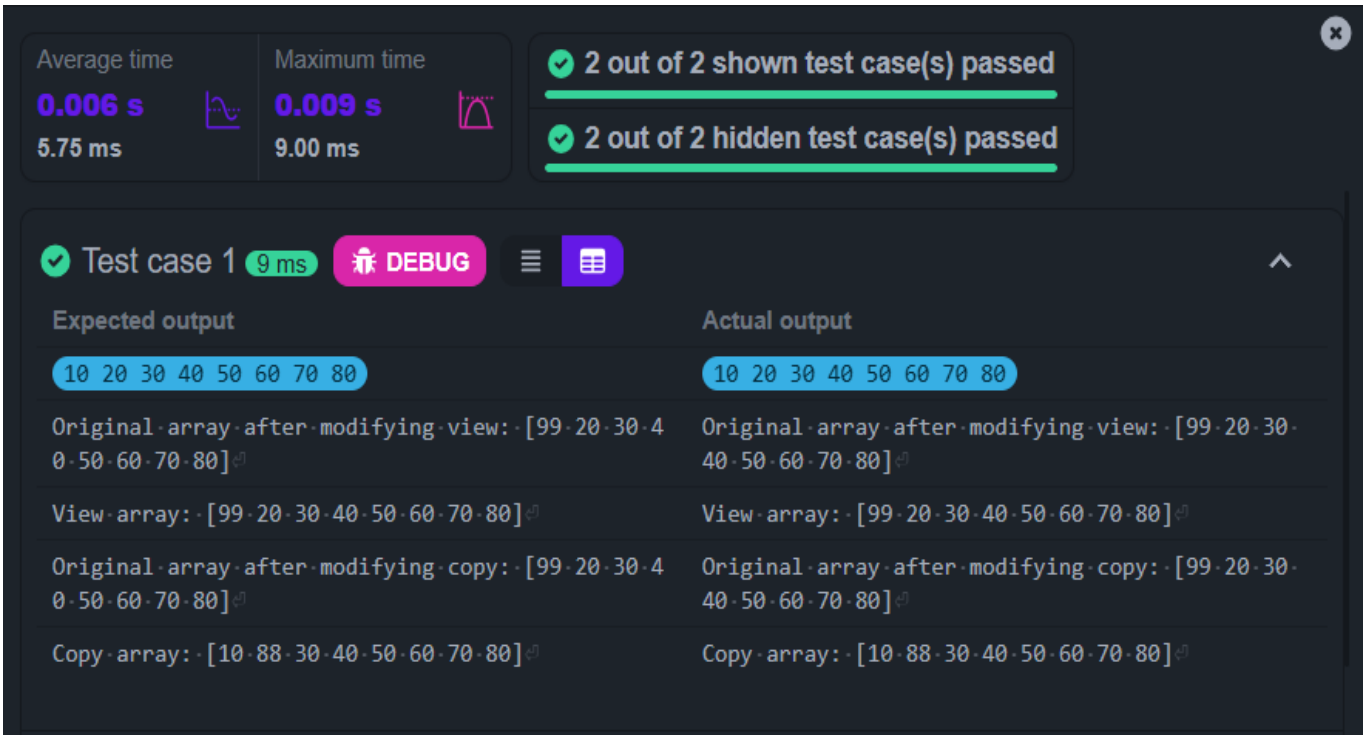
- A single line of space-separated integers.

Output Format:

- After modifying the view:
  Original array after modifying view: <original_array>
  View array: <view_array>
- After modifying the copy:
  Original array after modifying copy: <original_array>
  Copy array: <copy_array>

```
import numpy as np

inputlist = list(map(int,input().split(" ")))

original_array = np.array(inputlist)

view_array =original_array.view()

copy_array =original_array.copy()

view_array[0] = 99

print("Original array after modifying view:", original_array)

print("View array:", view_array)
```

```
copy_array[1] = 88

print("Original array after modifying copy:", original_array)

print("Copy array:", copy_array)
```



3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, **array1**, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- **search_value**: The value to search for in the array.
- **count_value**: The value to count its occurrences in the array.
- **broadcast_value**: The value to add for broadcasting across the

array.

You need to complete the code to perform the following operations:

1. **Searching**: Find the indices where **`search_value`** appears in **`array1`** and print these indices.

2. **Counting**: Count how many times **`count_value`** appears in **`array1`** and print the count.

3. **Broadcasting**: Add **`broadcast_value`** to each element of **`array1`** using broadcasting, and print the resulting array.

4. **Sorting Input Format:**

1. A single line containing space-separated integers representing **`array1`**.
2. An integer **`search_value`** represents the value to search for in the array.
3. An integer **`count_value`** represents the value to count in the array.
4. An integer **`broadcast_value`** represents the value to add to each element of the array.

Output Format:

1. The indices where **`search_value`** occurs in **`array1`**.
2. The count of occurrences of **`count_value`** in **`array1`**.
3. The array after adding the **`broadcast_value`** to each element.
4. The sorted array.

```
import numpy as np
```

```
array1 = np.array(list(map(int, input().split())))
```

```
search_value = int(input("Value to search: "))
```

```
count_value = int(input("Value to count: "))
```

```
broadcast_value = int(input("Value to add: "))
```

```
indices=np.where(array1==search_value)[0]

print(indices)

count= np.count_nonzero(array1==count_value)

print(count)

broadcasted_array= array1+ broadcast_value

print(broadcasted_array)

sorted_array= np.sort(array1)

print(sorted_array)
```

3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.

- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details

print("All student Details:\n", a )

# 2. print total students

print("Total Students:", a.shape[0])

# 3. Print all student Roll numbers

print("All Student Roll Nos",a[:,0]   )

# 4. Print subject 1 marks

print("Subject 1 Marks", a[:,1])

# 5. print minimum marks of Subject 2
```

```
print("Min marks in Subject 2", np.min(a[:,2])     )

# 6. print maximum marks of Subject 3

print("Max marks in Subject 3", np.max(a[:,3])    )

# 7. Print All subject marks

print("All subject marks:", a[:, 1:]      )

# 8. print Total marks of students

print("Total Marks", np.sum(a[:,1:],axis=1)       )

# 9. print average marks of each student

avg= np.mean(a[:,1:],axis=1)

print(np.round(avg,1))

# 10. print average marks of each subject

print("Average Marks of each subject", np.mean(a[:,1:],axis=0)       )

# 11. print average marks of S1 and S2

print("Average Marks of S1 and S2", np.mean(a[:, 1:3],axis=0)         )

# 12. print average marks of S1 and S3

print("Average Marks of S1 and S3", np.mean(a[:,[1,3]],axis=0)              )

# 13. print Roll number who got maximum marks in Subject 3

print("Roll no who got maximum marks in Subject 3", a[np.argmax(a[:, 3]),0])

# 14. print Roll number who got minimum marks in Subject 2

print("Roll no who got minimum marks in Subject 2", a[np.argmin(a[:, 2]),0])
```

```
# 15. print Roll number who got 24 marks in Subject 2

print("Roll no who got 24 marks in Subject 2", a[a[:, 2]==24,0:1] )

# 16. print count of students who got marks in Subject 1 < 40

print("Count of students who got marks in Subject 1 < 40", np.sum(a[:, 1]<40)            )

# 17. print count of students who got marks in Subject 2 > 90

print("Count of students who got marks in Subject 2 > 90:", np.sum(a[:,2]>90)           )

# 18. print count of students in each subject who got marks >= 90

print("Count of students in each subject who got marks >= 90:", np.sum(a[:,1:]>=90,axis=0)            )

# 19. print count of subjects in which each student got marks >= 90

print("Roll no:",a[:,0]      )

print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:]>=90,axis=1)       )

# 20. Print S1 marks in ascending order

print(np.sort(a[:,1]))



# 21. Print S1 marks >= 50 and <= 90

print(a[(a[:,1]>=50)&(a[:,1]<=90)])

print(a)

# 22. Print the index position of marks 79

print(np.where(a[:,1]==79))
```

Average time
0.022 s
22.00 ms

Maximum time
0.022 s
22.00 ms

1 out of 1 shown test case(s) passed

Test case 1 22 ms DEBUG

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]